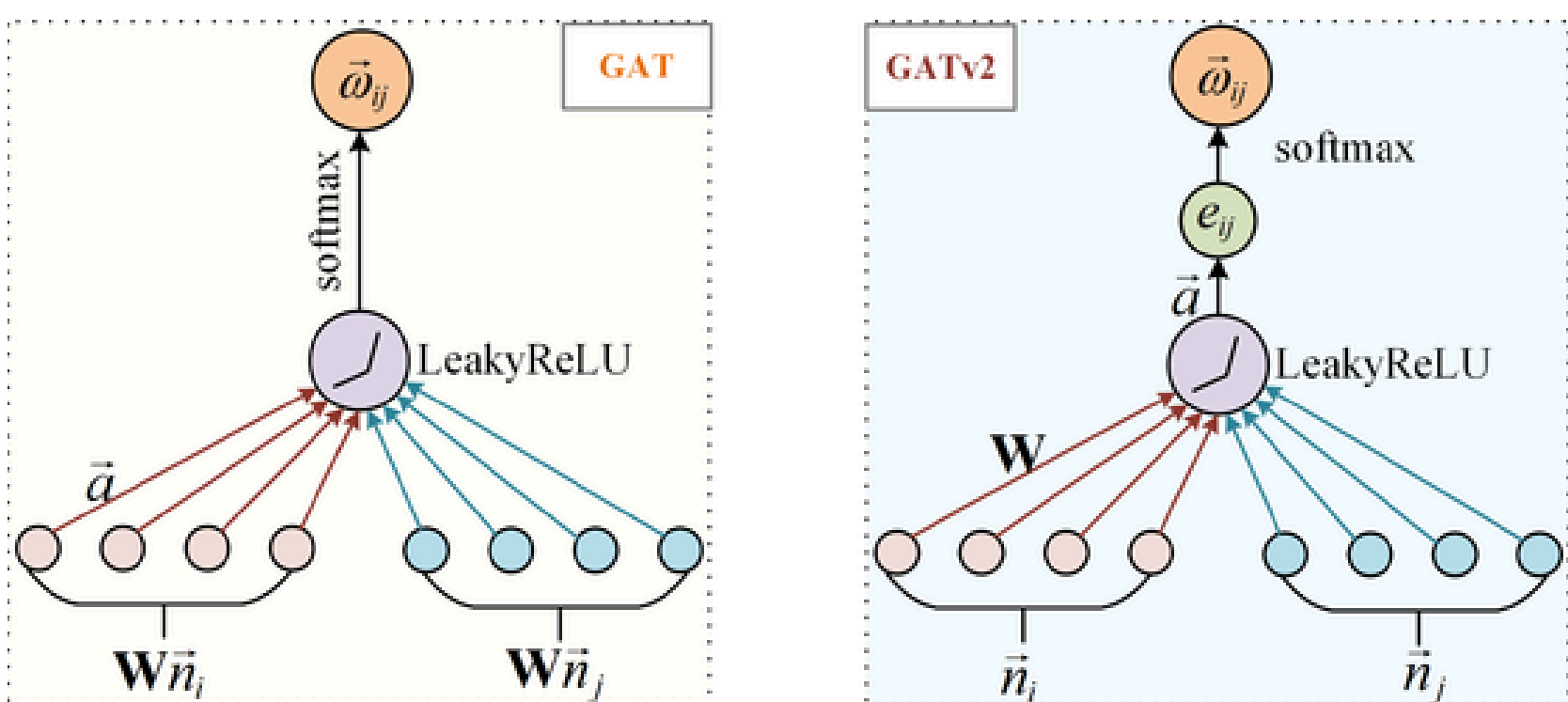


How Attentive are Graph Attention Networks?

We explore the limitations of standard Graph Attention Networks (GAT) in modeling complex and query-dependent relationships within graphs. While GAT was designed to allow nodes to attend to their neighborhoods, its architecture suffers from "static attention," meaning the ranking of neighbor importance is global.



Hypothesis: By utilizing dynamic attention, GATv2 computes a unique, query-dependent ranking of neighbors. This flexibility allows the model to filter out structural noise and irrelevant edges that standard GATs—limited by static rankings—are forced to attend to.

GAT (Veličković et al., 2018): $e(\mathbf{h}_i, \mathbf{h}_j) = \text{LeakyReLU}(\mathbf{a}^\top \cdot [\mathbf{W}\mathbf{h}_i || \mathbf{W}\mathbf{h}_j])$

GATv2 (our fixed version): $e(\mathbf{h}_i, \mathbf{h}_j) = \mathbf{a}^\top \text{LeakyReLU}(\mathbf{W} \cdot [\mathbf{h}_i || \mathbf{h}_j])$

We will evaluate the relative performance of GATv2 against standard GAT architectures across tasks requiring high selectivity.

Why GATs?

Applications:

Explainable AI, Fault Detection, Recommendation + Search Systems, Drug Discovery



- GCNs:** Treat all neighbor nodes equally during aggregation (isotropic), limiting their expressive power on complex graphs.
- GATs:** Solved this by assigning different weights to neighbors based on node features, but they suffer from "static attention" (a fixed, global ranking of neighbors).
- GATv2:** Introduces dynamic attention, computing unique, query-dependent weights to effectively filter out structural noise and isolate relevant edges.

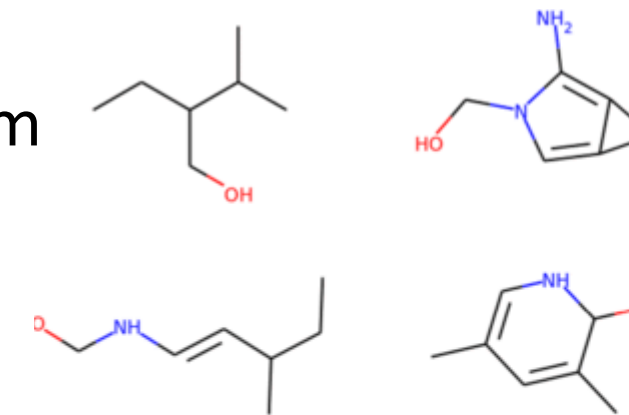
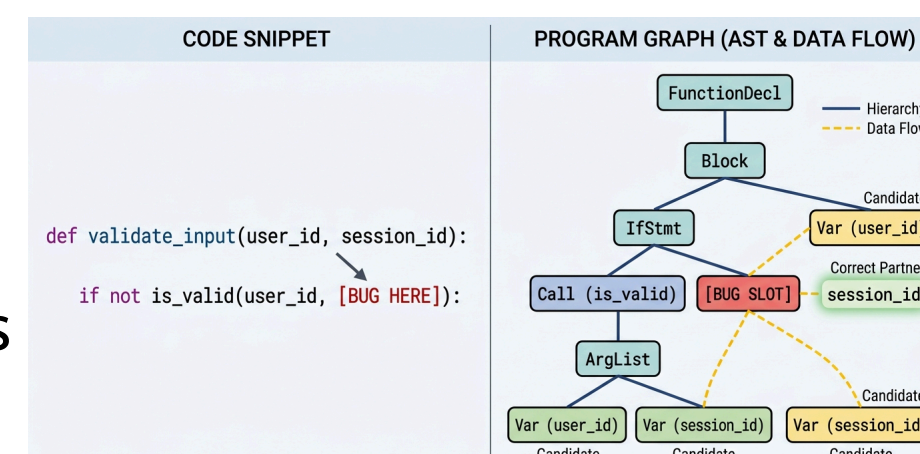
Methodology

VARMISUSE Dataset: Computer programs $\rightarrow$ Graphs (ASTs), represent syntactic/semantic interactions in code

Synthetic Data: Constructed "Dictionary" graphs with various levels of neighbor interference (k) or random noisy edge injection to isolate specific failure modes of GAT's static attention.

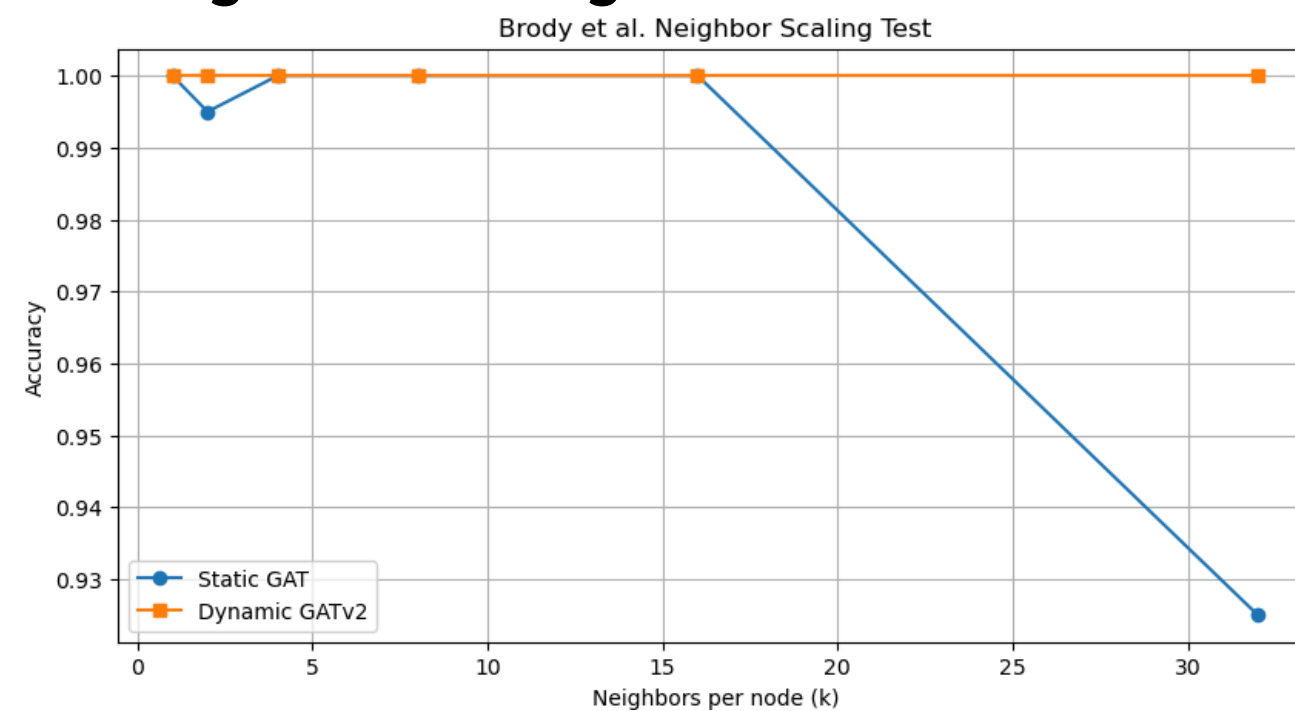
QM9 Dataset: 134K organic molecules, providing high-fidelity quantum chemical properties. Evaluates ability of models to predict molecular properties based on a molecule's geometric/atomic structure.

Models/Tools: PyTorch NNs, Matplotlib, Pytorch GNN's. Trained on Google Colab T4 and A100 GPUs.

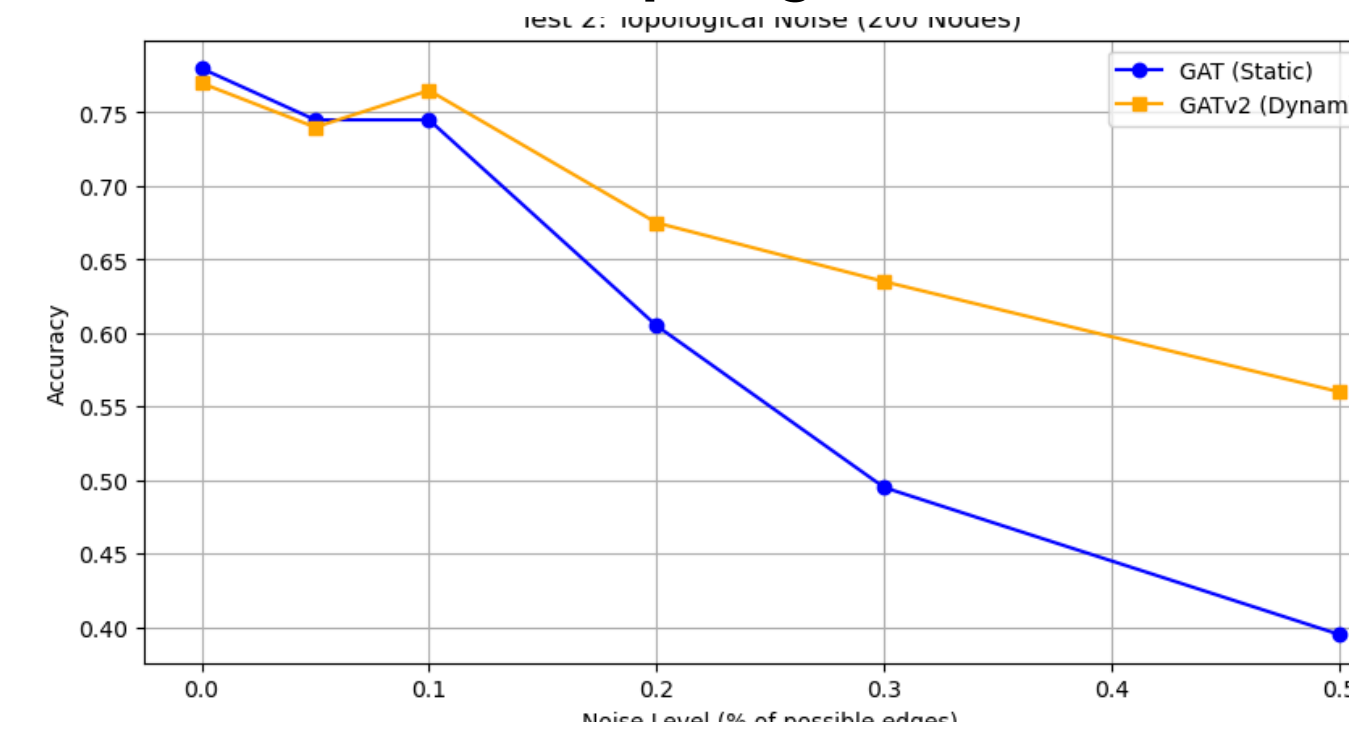


Results

Neighbor Scaling Test

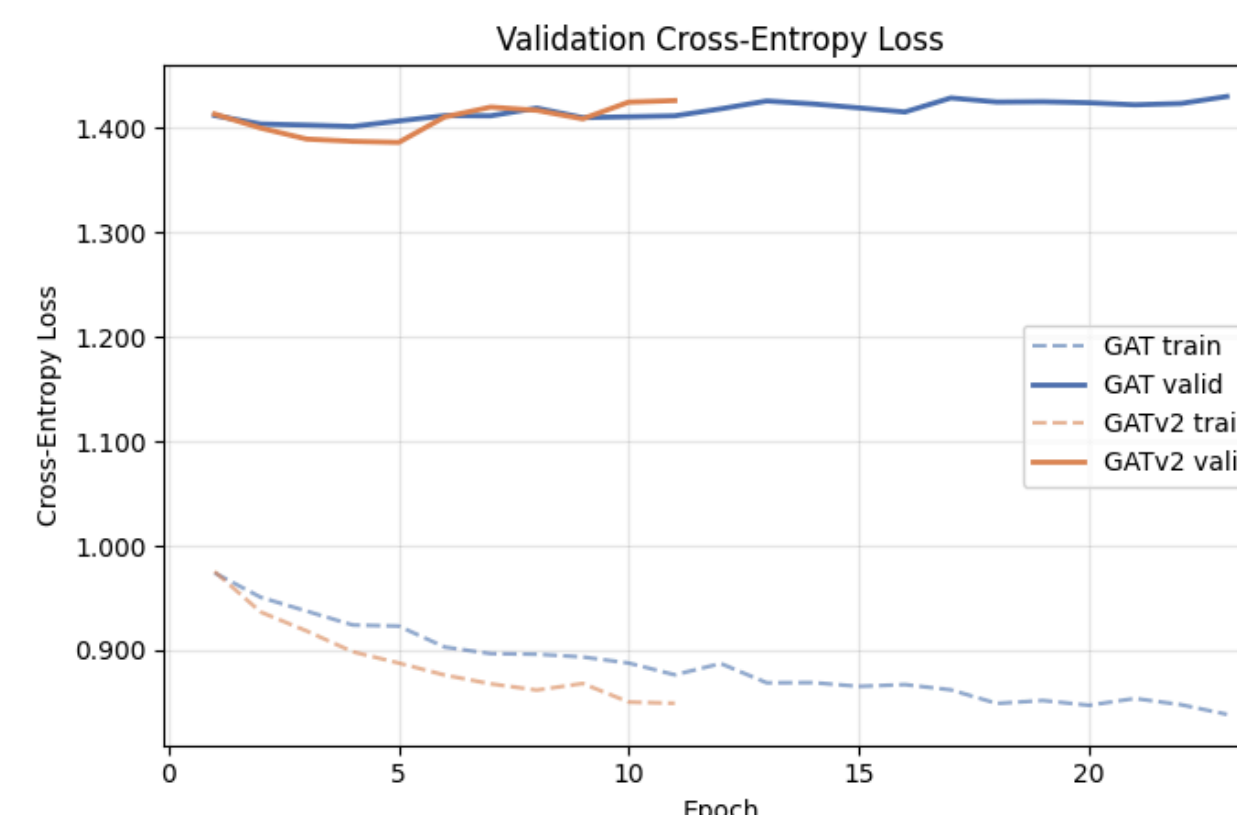
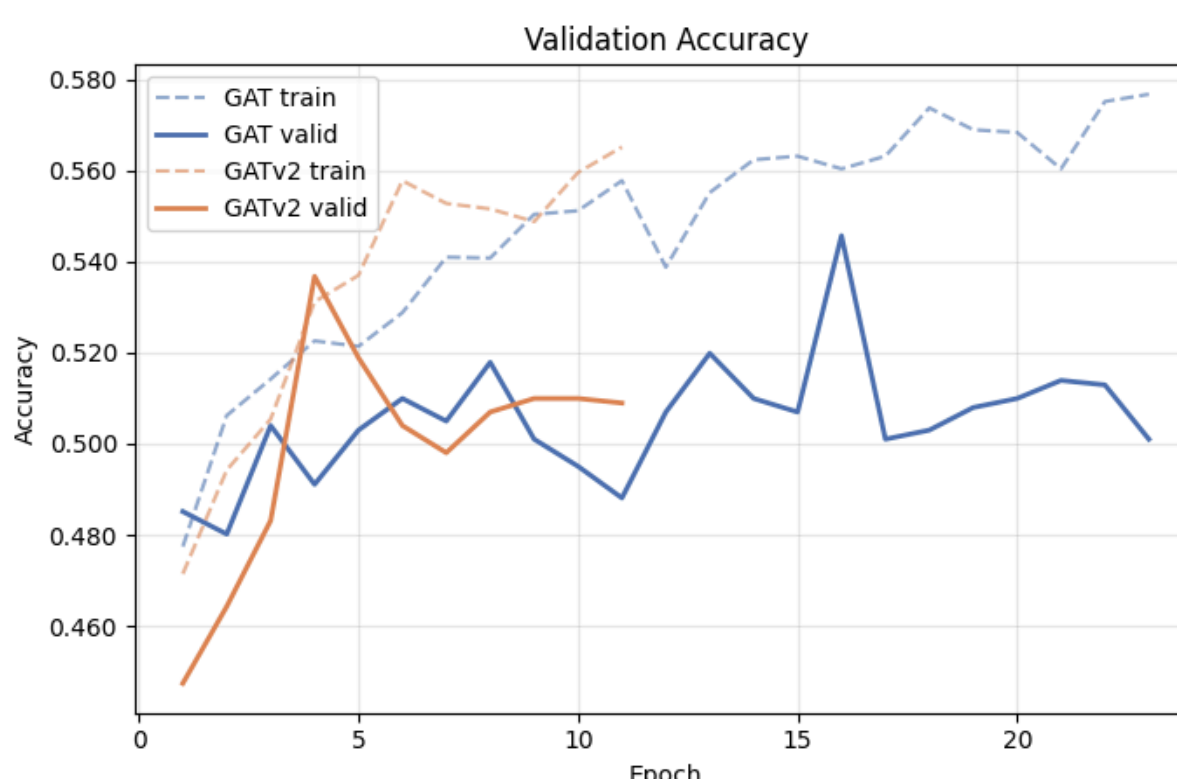


Robustness to Topological Noise



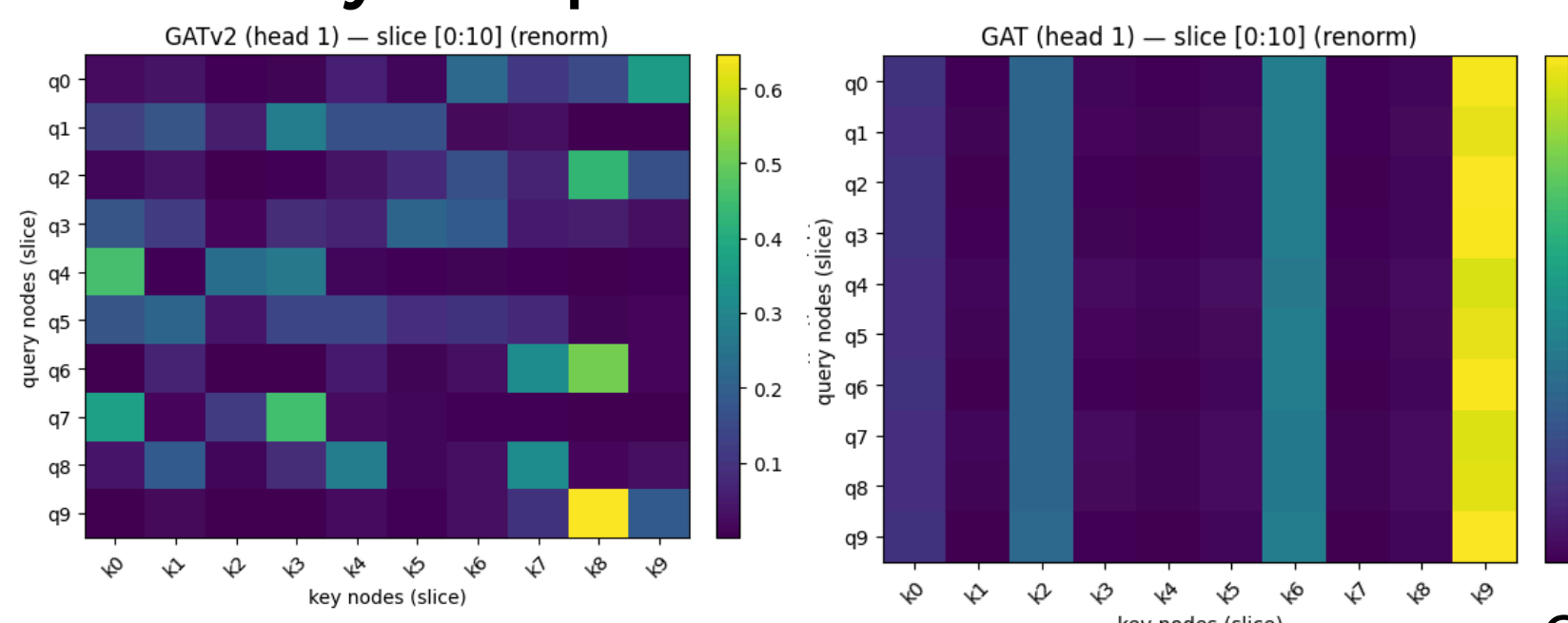
GATv2 filters out irrelevant edges, leading to stable performance. GAT suffers because it cannot ignore globally high-ranked noisy neighbors.

VARMISUSE Dataset

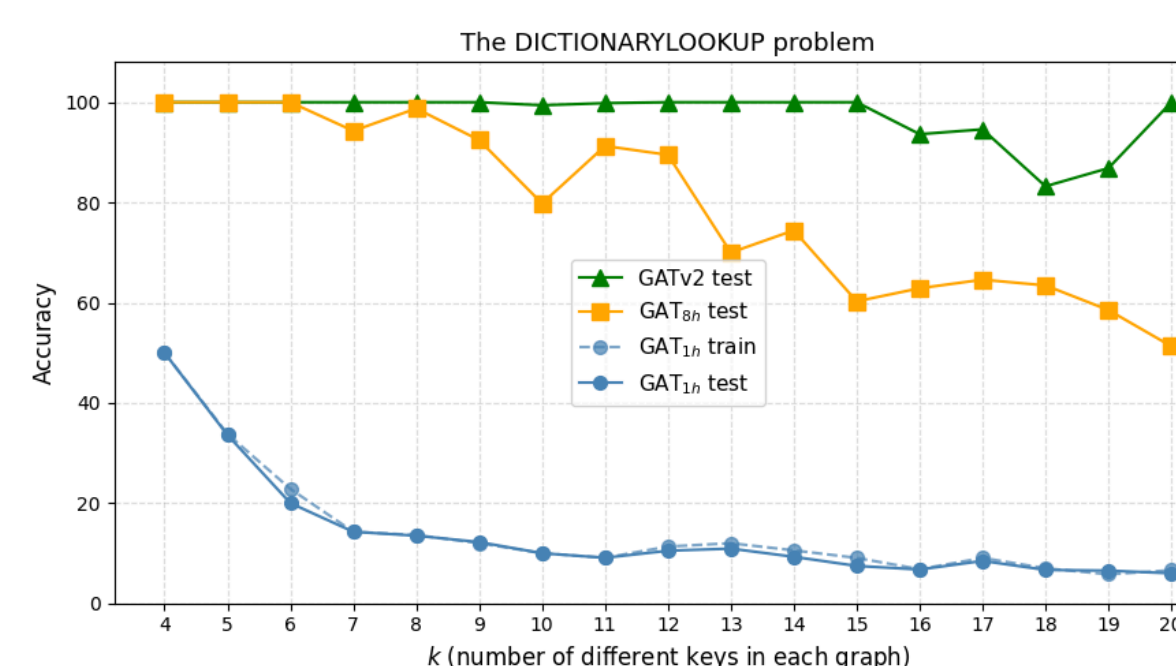


GATv2 effectively filters out irrelevant syntactic noise, isolating the specific semantic relationships needed to detect variable misuse

Dictionary Lookup



Attention Weight Matrix

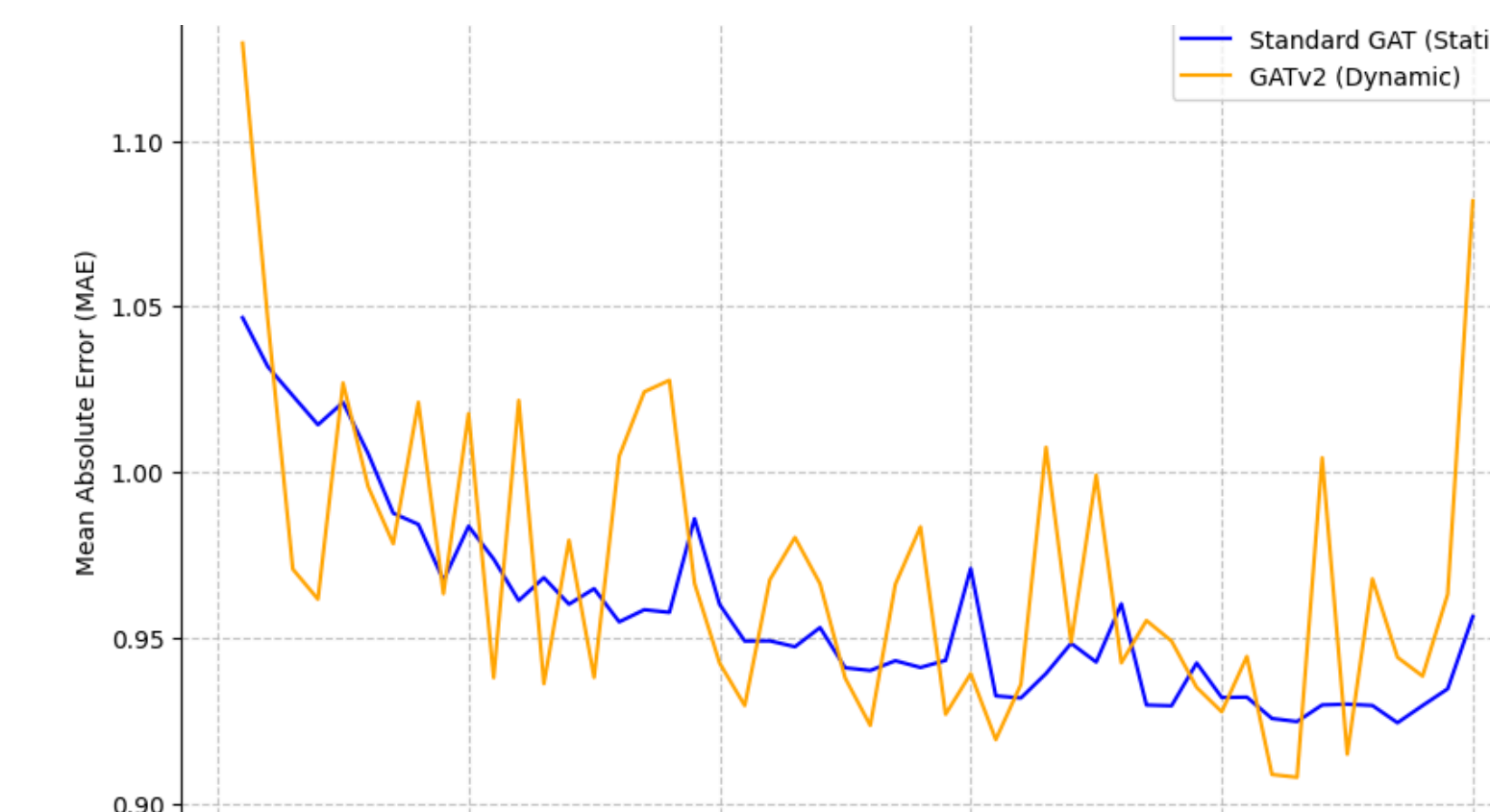


GATv2 outperforms due to selective key choices

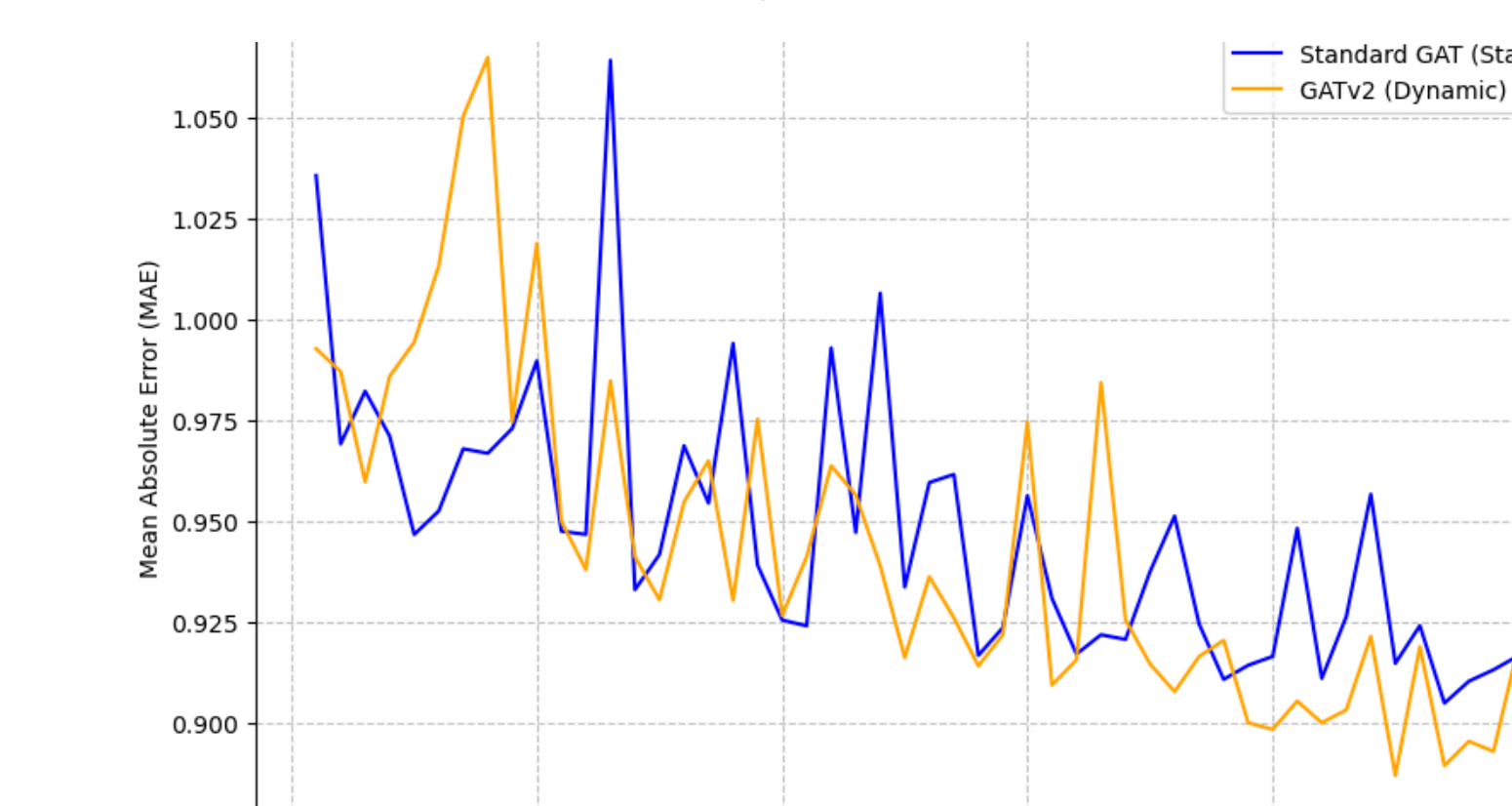
Additional Experiment

QM9 Dipole Moment Prediction

With 4 heads, hidden layers = 8



With 4 heads, hidden layers = 32



GAT's 'global look' provides a stronger inductive bias for small molecules. GATv2's ability to look 'deeper' into specific node pairs introduces unnecessary complexity for the relatively uniform chemical environments in QM9

Conclusion

Across both synthetic and real-world datasets, our evaluation confirms that **GATv2's** dynamic attention mechanism successfully overcomes the 'static attention' bottleneck of standard **GATs**. By computing unique, query-dependent neighbor rankings, GATv2 consistently filters out structural noise, leading to **superior robustness** and **accuracy** on complex relational tasks.

Future Work

- OGB Integration:** Scale up our evaluation by benchmarking GATv2 against standard GAT on large-scale OGB datasets
- Expanded Benchmarking:** Compare GATv2's performance against other standard graph architectures, such as GCNs (isotropic aggregation) and GINs (maximum expressive power)

References

- How Attentive are Graph Attention Networks <https://arxiv.org/pdf/2105.14491>
- Graph Attention Networks <https://arxiv.org/abs/1710.10903>

